NEPSE Forecasting Study — Technical Documentation

Nov 8, 2026 · @Rawadeep

Summary

After cleaning 91 million trades and passing a ten-point leakage audit, the calibrated ensembles forecast next-session direction better than chance on the untouched test (59.0% against a 55.3% majority baseline). At 3, 5 and 10 sessions accuracy cannot be told apart from the baseline, and 70% accuracy appears only on a tiny share of calls.

- **Data.** 91,048,648 trades parsed and 91,009,140 kept (99.957%), after removing 39,500 exact duplicates. Fourteen months of thin floorsheet coverage are handled by taking prices from the OHLCV table. Two sources, the nine single-quarter fundamentals tables and the feature store, were rejected because they leak future information.
- **Accuracy.** At 1 session, 59.0% (95% interval 57.0–60.9%) against 55.3%, and the ensemble beat its baseline in all six out-of-fold folds. At 3, 5 and 10 sessions it scored 59.2%, 58.3% and 55.3% against baselines of 57.3%, 57.3% and 56.8%, none significant.
- **70% confidence.** At confidence ≥ 0.70 the 1-session ensemble was right 81.1% of the time, on 0.40% of stock-sessions (243 calls). The 95% interval, 63.3–85.2%, does not show that accuracy exceeds 70%, and nearly all of those calls are "down" in a falling market.
- **Trading.** The cross-sectional ranker has a clearly positive rank-IC (0.21–0.28 on the test), but long-only portfolios earn no net return distinguishable from zero after costs, and the ranker's top-10 portfolio loses money at every horizon.
- **What a manuscript can claim.** Calibrated probability skill at 1 and 3 sessions (test AUC 0.61, Holm $p < 0.001$ and $p = 0.006$ against the base rate) and a ranker with positive rank-IC. It cannot claim 70% accuracy, and it should say that the numbers come from scikit-learn stand-ins, not from CatBoost, LightGBM, XGBoost, the Transformer or the TFT, which could not be installed here. Those five families have since been run on a separate 100-stock lite panel; that is a different experiment and none of its figures appear in this document. See "Experiment register and run lineage".

Data sources

The models use the three floorsheet dumps, the adjusted OHLCV table and the quarterly fundamentals table. The nine single-quarter fundamental tables and the feature store were audited and excluded.

File (table)	Contents	Rows	Period	Role
floorsheet_data_part1.zip	Trades: transaction no., symbol, buyer, seller, quantity, rate, amount, date	41,169,555	2018-10-01 – 2024-01-31	Used: broker features, raw prices, delisted stocks
floorsheet_data_p	Same schema	19,364,427	2024-02-01	Used

File (table)	Contents	Rows	Period	Role
art2.zip			– 2025-01-28	
floorsheet_data_part3.zip	Same schema	30,514,666	2025-02-02 – 2026-11-05	Used
ohlcv_data.sql (livedata_adjusted)	Daily adjusted OHLCV; NEPSE index and 16 sub-indices	540,011 (479 symbols)	1997-07-20 – 2026-11-05	Used: prices and labels for listed stocks, market features
fundamental_data.sql (new_fundamentals)	Quarterly EPS, P/E, P/B, ROE, D/E, revenue growth	5,625 reports (270 companies)	FY 2070/71 – 2082/83	Used: 7 point-in-time features
9 × fundamental_*.sql	Growth, valuation, leverage, liquidity, efficiency, profitability, risk, earnings quality, market performance	284 each	Latest quarter only	Excluded: one period per company; applying it to past dates leaks future data
nepse_prediction_featurestore.sql	Precomputed features and next-day target	148,289 (101 symbols)	2012-02-26 – 2026-11-04	Excluded: its fundamentals switch before the quarter ends (look-ahead)

All three zips pass CRC and schema checks; their SHA-256 hashes are in the pipeline's data report. Only the floorsheet contains duplicate keys (next section).

Cleaning and de-duplication

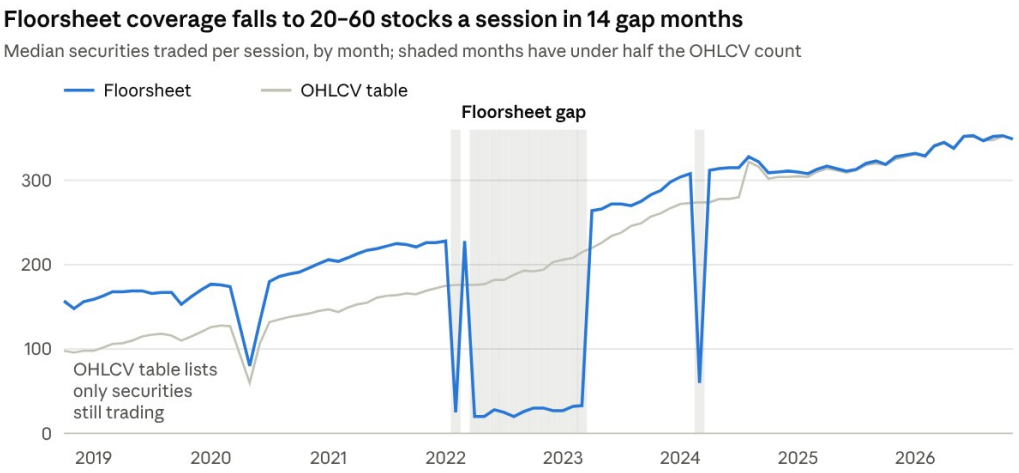
Of 91,048,648 trades, 91,009,140 (99.96%) survive cleaning: 39,500 duplicate transactions and 8 invalid rows are removed. De-duplication is global: each trade is filed under the month encoded in its transaction number, so every copy of a transaction meets in the same bucket regardless of which zip it came from.

Check	Rows	Action
Duplicate transaction number	39,500 (all exact copies, inside part 3, July 2026)	Removed
Quantity, rate or amount ≤ 0	8	Removed
Broker code `D01` (dealer account)	33,758	Kept as real trades (an earlier

Check	Rows	Action
		version wrongly dropped them)
Older transaction-number format, Oct–Dec 2018	102,076	Kept, flagged
Amount ≠ quantity × rate by more than NPR 1	710	Kept, flagged
`trade_time` empty	72,248,755 (79.4%)	Intraday order taken from the transaction number
Daily OHLC violations after aggregation	0	—

The clean trades aggregate to 405,370 stock-days (655 symbols, 1,840 sessions), and trade counts reconcile exactly with the daily totals. Storage drops from 7.3 GB of SQL to 2.9 GB of 32-byte binary records in monthly files; the job checkpoints every batch and resumes after an interruption.

The floorsheet itself has gaps. In Feb 2022, Apr 2022 – Mar 2023 and Mar 2024 it holds a median of 20–60 stocks per session, against 220–310 in complete months. The OHLCV table covers those months (190–290 symbols per session), so prices never depend on the gaps; broker features are left missing there rather than imputed.



floorsheet daily aggregates and livedata_adjusted · Oct 2018 – Nov 2026

Before 2023 the grey line runs below the blue one because the OHLCV table lists only securities still trading today; that is why the 82 delisted ordinary shares take their prices from the floorsheet.

Universe and prices

The study models 369 ordinary shares, giving 360,943 eligible stock-days across 362 stocks and 1,855 sessions (2018-10-01 – 2026-11-05). Promoter shares (143), debentures and fund series (112) and closed-end mutual funds (30) are excluded.

Security type	Rule	Count	Treatment
Ordinary share, listed	In the OHLCV table	287	Back-adjusted OHLCV prices
Ordinary share, delisted or merged (e.g. NCCB, NBB, MEGA, CBL, BOKL)	Only in the floorsheet	82	Floorsheet prices; kept to avoid survivorship bias
Promoter share	Listed symbol + "P" or "PO"	143	Excluded
Debenture or fund series	Digit or "/" in the symbol	112	Excluded
Closed-end mutual fund	Median price below NPR 30 (NAV ≈ NPR 10)	30	Excluded

Price construction:

- **Adjusted OHLCV is verified, not assumed.** Its close equals the raw floorsheet close within 0.5% on 95.9% of days since June 2026 and steps at bonus and rights dates, so it is back-adjusted. Volume is in raw shares.
- **Floorsheet prices of delisted stocks** are adjusted heuristically: a close-to-close move beyond 11% (the daily limit is ±10%) is treated as a corporate action, and that day's return is replaced by its open-to-close return.
- **Chaining.** 1,522 floorsheet bars that predate a stock's first OHLCV bar are rescaled to the adjusted basis at the first overlapping day. The scale sets only the price level; every feature and return is scale-free.
- **Repairs.** 19,433 OHLCV bars whose open or close lay outside the high–low range had high/low widened; 225 residual jumps over 11% among eligible stock-days were neutralised.
- **Eligibility** uses only the past: traded that session, at least 10 trading days in the last 20, a 60-session return available, and 29 earlier trading days for the sequence model. The session calendar is the union of OHLCV and floorsheet trading days; OHLCV history from June 2017 only warms up rolling windows.

Predictors

The models see 60 predictors per stock-day: 53 core features and 7 point-in-time fundamentals. Each uses only data available at the close of day t; the truncation test in the leakage audit confirms this. Missing values are left missing in the data: gradient boosting splits on them directly; ExtraTrees and logistic regression use training-median imputation; the neural networks use median imputation plus two flags (floorsheet missing, fundamentals missing).

Notation: P = adjusted close; O, H, L = same-day open, high, low; r = daily return; V = share volume; A = floorsheet turnover (NPR); N = number of trades; k = window in sessions. Source: Bars = adjusted OHLCV (floorsheet bars for delisted stocks); FS = floorsheet; IDX = NEPSE index; FUND = quarterly reports.

Feature	Group	Definition	Source	Missing %
ret1, ret2, ret3, ret5, ret10, ret20, ret60	Return (7)	P(t) / P(t−k) − 1 for k = 1, 2, 3, 5, 10, 20, 60	Bars	0.0
vol5	Volatility	Std. dev. of r over 5 sessions (min. 3 trades)	Bars	0.1
vol10	Volatility	Std. dev. of r over 10 sessions	Bars	0.2
vol20	Volatility	Std. dev. of r over 20 sessions	Bars	0.2
vol60	Volatility	Std. dev. of r over 60 sessions	Bars	0.4
hl_range	Candle	(H − L) / close	Bars	0.0
oc_ret	Candle	close / O − 1, clipped to ±20%	Bars	0.0
close_vwap	Candle	Floorsheet close / floorsheet VWAP − 1	FS	11.5
gap	Candle	O(t) / P(t−1) − 1	Bars	0.0
close_pos	Candle	(close − L) / (H − L); 0.5 when H = L	Bars	0.0
ma5_dist, ma10_dist, ma20_dist, ma50_dist	Trend (4)	P / simple moving average over k − 1	Bars	0.0
rsi14	Oscillator	100 × mean gain / (mean gain + mean loss), 14 sessions	Bars	0.0
macd_hist	Oscillator	(EMA12 − EMA26 − its EMA9) / P	Bars	0.0
bb_pos	Oscillator	(P − SMA20) / (2 × SD20)	Bars	0.0
atr14_pct	Oscillator	Mean true range over 14 sessions / P	Bars	0.1
stoch_k14	Oscillator	(P − lowest L) / (highest H − lowest L), 14 sessions	Bars	0.1
drawdown60	Range	P / highest P over 60 sessions − 1	Bars	0.0
dist_high252	Range	P / highest P over 252 sessions − 1 (min. 120)	Bars	3.6

Feature	Group	Definition	Source	Missing %
dist_low252	Range	P / lowest P over 252 sessions − 1 (min. 120)	Bars	3.6
volratio_1_20	Liquidity	V(t) / mean V over 20 sessions	Bars	0.0
volratio_5_20	Liquidity	Mean V over 5 / mean V over 20	Bars	0.0
log_amount	Liquidity	ln(1 + A)	FS	11.5
log_trades	Liquidity	ln(1 + N)	FS	11.5
amihud20	Liquidity	ln(1 + 10^9 × mean of |r| / A over 20 sessions)	FS	9.2
stale20	Liquidity	Share of the last 20 sessions with no trade or zero return	Bars	0.0
turnover_trend	Liquidity	Mean A over 5 / mean A over 20	FS	10.6
avg_trade_size	Liquidity	ln(shares traded / N)	FS	11.5
trades_ratio20	Liquidity	N / mean N over 20 sessions	FS	11.8
buyer_hhi, seller_hhi	Broker	Herfindahl index of buying (selling) brokers' volume shares	FS	11.5
buyer_top1, seller_top1	Broker	Largest single buying (selling) broker's volume share	FS	11.5
buyer_top5, seller_top5	Broker	Top-5 buying (selling) brokers' volume share	FS	11.5
log_buyers, log_sellers	Broker	ln(1 + number of distinct buying (selling) brokers)	FS	11.5
self_share	Broker	Share of volume where buying broker = selling broker	FS	11.5
large_share	Broker	Share of volume in trades of 1,000 shares or more	FS	11.5
size_cv	Broker	Coefficient of variation of trade size	FS	11.5
broker_imbalance	Broker	buyer_top5 − seller_top5	FS	11.5
buyer_hhi_chg5	Broker	buyer_hhi − its 5-session mean	FS	11.8
mkt_ret1	Market	NEPSE index daily return	IDX	0.0

Feature	Group	Definition	Source	Missing %
		(equal-weighted universe on the 8 sessions without an index value)		
mkt_vol20	Market	Std. dev. of mkt_ret1 over 20 sessions	IDX	0.0
rel_ret5	Market	ret5 − NEPSE 5-session return	Bars, IDX	0.0
fund_ep	Fundamental	Earnings yield: annualised EPS / quarter-end price × P(quarter end) / P(t)	FUND, Bars	27.6
fund_bp	Fundamental	Book-to-price: (1 / P/B at quarter end) × P(quarter end) / P(t)	FUND, Bars	21.1
fund_roe	Fundamental	Return on equity (%), as reported	FUND	31.0
fund_de	Fundamental	Debt-to-equity, as reported	FUND	31.7
fund_rev_growth	Fundamental	Revenue growth (%), as reported	FUND	34.6
fund_eps_yoy	Fundamental	(EPS − EPS of the same quarter a year earlier) / |that EPS|, clipped to ±5	FUND	41.5
fund_age	Fundamental	ln(1 + sessions since the report became usable)	FUND	16.1

Floorsheet features are missing on 11.5% of stock-days, almost all in the floorsheet gap months. A report is usable from quarter end + 45 days and is dropped after 260 sessions without a newer one; P/E and P/B in the table were computed with the quarter-end price (86% of reports match it within 2%), so valuation ratios are rolled forward with the stock's own adjusted return and never use a price after day t.

Targets

Each horizon h ∈ {1, 3, 5, 10} sessions has a binary label: 1 if the adjusted close h sessions ahead is above today's close, 0 if below. Stock-days whose price is exactly unchanged are left unlabelled (5.6% at 1 session, 0.5% at 10), because a tie is neither direction.

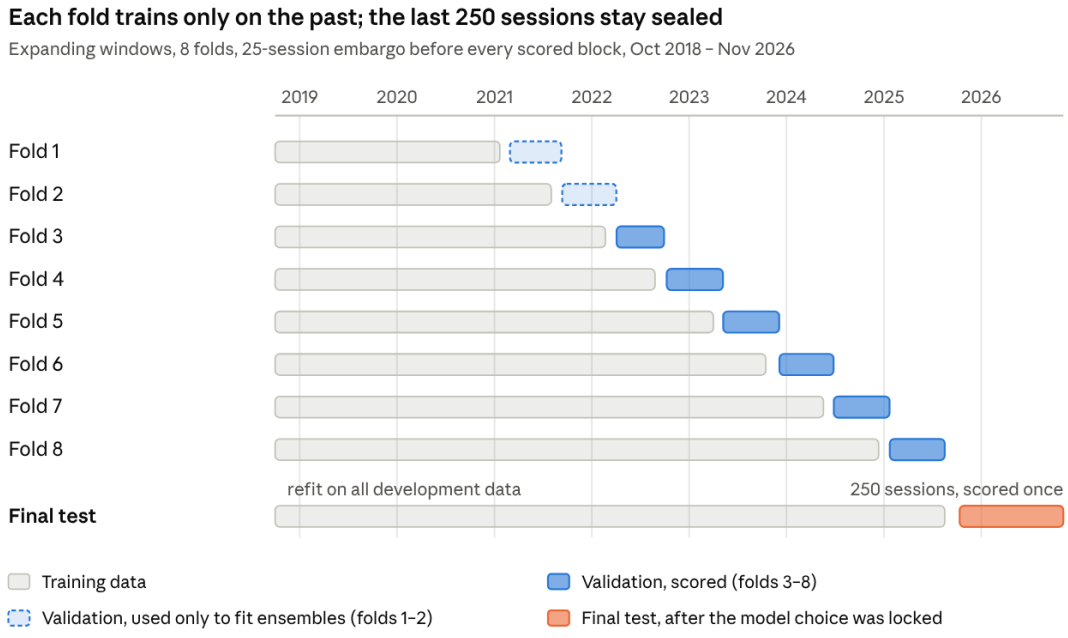
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y_{i,t}^{(h)} = \mathbf{1}\left[\frac{P_{i,t+h}}{P_{i,t}} - 1 > 0\right], \
quad \text{undefined if } P_{i,t+h} = P_{i,t}
```

Horizon	Labelled stock-days	Share up	Unchanged, excluded	Mean forward return
1 session	340,322	45.5%	5.6%	0.08%
3 sessions	354,301	45.7%	1.6%	0.23%
5 sessions	355,809	46.2%	1.0%	0.38%
10 sessions	356,498	46.6%	0.5%	0.77%

Fewer than half the labels are "up", so always predicting "down" already scores 53–55%; every accuracy figure below is compared with that majority baseline in its own test window. The backtest uses a separate tradable return: buy at the next session's open, sell at the close h sessions after today, which removes the closing-price bounce from the profit calculation.

Validation design

Every score in this document is out-of-sample in time. Fold k trains on all stock-days before its block, minus a 25-session embargo (longer than the 10-session label), and is scored on the next 131 sessions (134 in fold 8).



walk-forward validation · 8 folds + final test

Folds 1–2 only feed the ensemble and calibration layers, so every out-of-fold number comes from folds 3–8 (789 sessions). The final test was scored once, after the model choice per horizon was locked and hashed.

Block	Training window	Scored window	Training stock-days	Scored stock-days
Fold 1	2018-10-01 – 2021-01-21	2021-02-28 – 2021-09-09	72,100	23,150
Fold 2	2018-10-01 – 2021-08-02	2021-09-12 – 2022-04-03	94,897	23,258
Fold 3	2018-10-01 – 2022-02-21	2022-04-04 – 2022-09-29	118,468	21,146
Fold 4	2018-10-01 – 2022-08-26	2022-10-09 – 2023-05-08	139,758	23,332
Fold 5	2018-10-01 – 2023-04-02	2023-05-09 – 2023-12-05	162,246	27,599
Fold 6	2018-10-01 – 2023-10-16	2023-12-06 – 2024-06-26	189,215	31,235
Fold 7	2018-10-01 – 2024-05-19	2024-06-27 – 2025-01-22	219,886	31,781
Fold 8	2018-10-01 – 2024-12-12	2025-01-23 – 2025-08-17	251,706	32,629
Final test	2018-10-01 – 2025-08-17 (refit)	2025-10-12 – 2026-11-05	290,409	64,387

Leakage audit

All 10 checks pass. The strongest is the truncation test: every source was cut at 2024-06-02 (fundamentals by their usable date), the panel was rebuilt, and all 60 features came out identical on all 263,496 earlier stock-days. A centred window, a back-fill or a fundamental used before publication would have changed them.

Check	What it guards against	Result
Embargo covers the label horizon	A training label that resolves inside the validation block	Pass: 25-session embargo, 10-session labels, 16 sessions to spare
Validation blocks tile the period	Overlapping or missing evaluation blocks	Pass: 8 contiguous blocks
First training window	Too little history in fold 1	Pass: 504 sessions
Final test isolated	Development data touching the test	Pass: 25-session gap; test = 250 sessions, 2025-10-12 – 2026-11-05
Development and test disjoint	A stock-day in both sets	Pass: 290,409 vs 64,387 rows, no overlap
No duplicate stock-sessions	Double-counted rows	Pass
No labels at the series end	Labels built from missing future prices	Pass: 0 labels in the last h sessions

Check	What it guards against	Result
Truncation: same rows	Eligibility depending on future data	Pass: 263,496 / 263,496 rows
Truncation: same features	Any feature using future data	Pass: max difference 0, 0 missing-value mismatches
No suspicious correlation	A feature that is secretly the label	Pass: highest |correlation| with the 1-session label is 0.041 (ret2)

Source-level checks (stage 01b) come first: they rejected the feature store, whose fundamentals switch about 30 days before quarter end, and the nine single-quarter snapshot tables.

Models

The results in this document come from five scikit-learn classifiers and one ranker, each the algorithm its name says. CatBoost, LightGBM, XGBoost, the Transformer and the TFT are implemented in the same pipeline (`--family native`) but could not be installed in this environment, so they did not produce any number in this document. They were later run on a reduced 100-stock panel in a separate workspace, recorded under "Experiment register and run lineage"; results from that run belong to it, not to the tables here.

Model	Type	Settings	Training rows per fit	Outputs
HistGradientBoosting	Gradient-boosted trees	Learning rate 0.05, 31 leaves, min. leaf 200, L2 = 5, 80% of features per split, up to 1,000 trees; stops after 50 rounds without improvement on the last 12% of training sessions (15-session purge)	All	One model per horizon
ExtraTrees	Randomised tree ensemble	200 trees, min. leaf 100, 30% of features per split, 50% bootstrap; median imputation	Up to 150,000 (random sample)	One model per horizon
Logistic regression	Linear	L2 penalty, C = 0.1; quantile-to-normal transform; median imputation	All	One model per horizon

Model	Type	Settings	Training rows per fit	Outputs
MLP	Neural network	Hidden layers 128 and 64, ReLU, Adam (lr 0.001, batch 512, L2 0.001); early stopping, patience 4, at most 40 epochs; median/IQR scaling clipped at ±5 plus 2 missingness flags	Up to 150,000	One network, 4 horizon outputs
Windowed MLP (sequence)	Neural network on a 20-session window	Same network; inputs = 12 key features over the previous 19 sessions + the current 60 features (290 inputs)	Up to 120,000	One network, 4 horizon outputs
Ranker	Gradient-boosted regression on ranks	Target = within-session quintile (0–4) of the forward return, sessions with at least 10 stocks; lr 0.05, 31 leaves, up to 400 trees, early stopping	All	One model per horizon; used for ranking (IC), not in the probability ensemble

Every fit sees only stock-days up to its fold's training end. Time order is never shuffled, except inside the MLP's own early-stopping hold-out, which stays within the training period.

Ensembles and calibration

The five classifiers' probabilities are combined four ways, and every combination and base model is also calibrated two ways, giving 12 ensemble candidates per horizon (27 with the calibrated base models). Nothing is fitted on the rows it is scored on: each layer learns only from earlier folds.

Ensemble	Rule
Average	Mean of the five probabilities
Weighted	Non-negative weights summing to 1 that minimise log-loss on earlier folds
Stacking	Logistic regression on the five log-odds, fitted on earlier folds
Regime-aware stacking	A separate stacker for each of 4 market regimes (NEPSE 60-session return up/down × 20-session volatility above/below its median), falling back to the global stacker when a regime has fewer than 3,000 rows

```
p_{\text{stack}} = \sigma\!\Big(\beta_0 + \sum_{m=1}^{5} \beta_m \,\
operatorname{logit} p_m\Big), \qquad p_{\text{Platt}} = \sigma\big(a + b \
operatorname{logit} p\big)
```

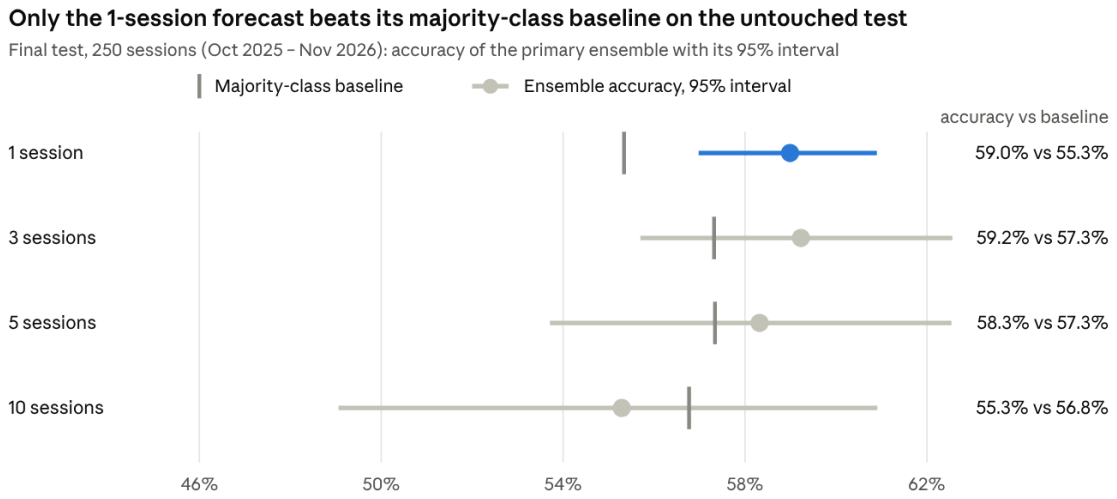
Isotonic calibration fits a monotone step function (pool-adjacent-violators), clipped to [0.005, 0.995].

Nested walk-forward order:

1. Base models produce out-of-fold probabilities for folds 1–8.
2. For fold k, ensemble weights and stackers are fitted on folds before k (minus the 25-session embargo), giving ensemble outputs for folds 2–8.
3. For fold k, Platt and isotonic calibrators are fitted on ensemble outputs from folds 2 to k−1, giving calibrated outputs for folds 3–8.
4. All out-of-fold comparisons use folds 3–8 (Apr 2022 – Aug 2025), where every candidate exists.
5. For the final test, the candidate with the lowest out-of-fold log-loss is chosen per horizon and its name is hashed into a lock file before any test prediction is scored.

Prediction accuracy

On the untouched 250-session test, only the 1-session forecast beats the majority baseline with confidence: 59.0% against 55.3%, with a 95% interval of 57.0–60.9%. The 3-, 5- and 10-session forecasts cannot be told apart from always predicting the majority class.



stage 12 final test · primary ensemble (weighted) · 1,3,5,10-session horizons · scikit-learn model family

The majority class is "down" at every horizon, in the out-of-fold folds and on the test (1-session up-rate on the test: 44.7%), so the baseline is 54–57%, not 50%. Out-of-fold, the 1-session ensemble beat its baseline in all six scored folds; the 3-session ensemble beat it in one fold, and the 5- and 10-session ensembles in none.

Horizon	Out-of-fold accuracy vs baseline	Folds beating baseline	Out-of-fold AUC	Test AUC	Test ECE	p(test accuracy > baseline)
1 session	57.1% vs 54.8%	6 of 6	0.579	0.608	0.032	< 0.001

Horizon	Out-of-fold accuracy vs baseline	Folds beating baseline	Out-of-fold AUC	Test AUC	Test ECE	p(test accuracy > baseline)
3 sessions	54.1% vs 54.5%	1 of 6	0.535	0.610	0.049	0.144
5 sessions	52.8% vs 54.1%	0 of 6	0.525	0.597	0.062	0.317
10 sessions	52.3% vs 53.8%	0 of 6	0.521	0.584	0.073	0.685

Out-of-fold results cover folds 3–8 (789 sessions, 163,000–167,000 labelled rows per horizon). Test AUC is higher than out-of-fold AUC at every horizon. Either the final models had more training data or the test window was easier; one window cannot separate the two.

Model comparison

The ensembles were chosen on out-of-fold log-loss before the test was scored, and the weighted ensemble won at every horizon. On the test, a single HistGradientBoosting model has the highest AUC at 3 to 10 sessions, but it was not selected in advance, so it is shown here only for comparison.

Model	AUC, 1 session	AUC, 3 sessions	AUC, 5 sessions	AUC, 10 sessions
Weighted ensemble (primary)	0.608	0.610	0.597	0.584
Simple average	0.585	0.591	0.586	0.584
Stacking	0.607	0.604	0.590	0.580
Regime-aware stacking	0.595	0.577	0.545	0.536
HistGradientBoosting	0.606	0.629	0.607	0.609
ExtraTrees	0.586	0.583	0.582	0.586
Logistic regression	0.558	0.546	0.526	0.505
MLP	0.561	0.563	0.562	0.561
Windowed MLP	0.536	0.540	0.539	0.545

The tree models carry the signal and the neural networks add little. The two MLPs are badly calibrated before recalibration (ECE 0.11–0.24 on the test), which is why they gain the most from Platt and isotonic scaling. The full table for all 27 candidates at each horizon is in `research_report.md`.

Cross-sectional ranker

The ranker orders stocks within each session instead of calling direction. Its rank correlation with realised returns is clearly positive at every horizon, although it shrinks as the horizon lengthens.

Horizon	Out-of-fold rank-IC	Test rank-IC (HAC t)	Test top-minus-bottom quintile return	Test top-decile up-rate
1 session	0.206	0.284 (37.5)	1.02%	58.6%
3 sessions	0.173	0.270 (24.2)	1.45%	55.8%
5 sessions	0.142	0.248 (15.0)	1.67%	53.0%
10 sessions	0.105	0.213 (8.6)	2.04%	51.8%

The spread is a long-short quantity, and NEPSE retail investors can only go long. The Backtest section shows that the long-only version loses money after costs.

70% confidence

A 70% accuracy is reachable only by acting on a tiny share of calls. At confidence ≥ 0.70, the 1-session ensemble was right 81.1% of the time on the test, but that is 243 calls, 0.40% of all stock-sessions, and the 95% interval runs from 63.3% to 85.2%.

A call is made when the larger of P(up) and P(down) reaches the threshold. Each cell below shows accuracy, then coverage (the share of all stock-sessions that receive a call).

Horizon	≥ 0.55	≥ 0.60	≥ 0.65	≥ 0.70	Out-of-fold ≥ 0.70 [95% interval]
1 session	64.0% (50.1%)	69.6% (16.1%)	72.0% (2.9%)	81.1% (0.40%)	71.3% (1.2%) [62.3%, 78.3%]
3 sessions	66.5% (38.4%)	74.5% (7.2%)	83.6% (0.66%)	100% (0.03%, 22 calls)	68.5% (0.37%) [57.8%, 79.3%]
5 sessions	65.0% (25.8%)	64.3% (2.2%)	83.7% (0.08%, 49 calls)	no calls	49.4% (0.29%) [37.9%, 68.3%]
10 sessions	59.0% (28.5%)	62.9% (2.3%)	68.6% (0.06%, 35 calls)	1 call, wrong	53.3% (1.47%) [25.3%, 74.1%]

The first four columns are the untouched test. Four things limit what the table can support.

- **Almost every confident call is "down".** In the out-of-fold 1-session stack, over 99% of calls at ≥ 0.70 predicted a fall, in a market where 55% of outcomes were down anyway.
- **Small counts.** The 100% at 3 sessions rests on 22 calls, so its interval says nothing. The 5- and 10-session ensembles make almost no calls at ≥ 0.70, and none of them were accurate.
- **70% is not shown to be exceeded.** The one-sided block-bootstrap test of "accuracy above 70%" gives p = 0.407 out-of-fold and p = 0.149 on the test for 1 session, so neither rejects an accuracy of 70% or less.
- **Selection.** Four thresholds at four horizons are sixteen looks. The thresholds were fixed before scoring, but the best-looking cell is still the best of sixteen.

Calibration

The weighted ensemble is already close to calibrated before any rescaling: ECE is 0.008 out-of-fold and 0.032 on the test at 1 session. Platt scaling and isotonic regression, fit only on earlier folds, change little there (test ECE 0.027 and 0.021; log-loss 0.6720 to 0.6715 and 0.6707). At 5 and 10 sessions they pull every probability toward the base rate, so calibrated forecasts almost never pass 0.55 and every call defaults to the majority class. Their accuracy then equals the baseline exactly (57.3% and 56.8%), which is calibration working as intended on a model with little signal.

The test reliability curve shows the 1-session ensemble is slightly under-confident at the top: stock-sessions forecast at P(up) near 0.62 rose 77% of the time (982 observations). The curve for every horizon is in `research_report.md`.

Significance

The ensemble's probabilities beat the base rate at 1 and 3 sessions on the test, but not at 5 or 10 sessions, and its edge over a single model or a simple average is detectable only at 1 session.

Each cell is the mean log-loss advantage of the weighted ensemble (positive means the ensemble is better) with the Holm-corrected one-sided p-value from a Diebold–Mariano test on session-level losses, using a heteroskedasticity- and autocorrelation-consistent (HAC) variance whose lag is the horizon-dependent max(h, 5) + 5 sessions (10 for 1 to 5 sessions, 15 for 10).

Horizon	vs base rate (climatology)	vs best single model by out-of-fold loss (ExtraTrees)	vs simple average
1 session	+0.0145 (p < 0.001)	+0.0066 (p = 0.0002)	+0.0053 (p = 0.019)
3 sessions	+0.0105 (p = 0.006)	+0.0060 (p = 0.001)	+0.0039 (p = 0.87)
5 sessions	+0.0041 (p = 1.00)	+0.0032 (p = 0.35)	+0.0041 (p = 1.00)
10 sessions	−0.0004 (p = 1.00)	−0.0003 (p = 1.00)	+0.0082 (p = 1.00)

Three readings follow from the table and the out-of-fold tests.

- **Probability skill is wider than accuracy skill.** At 3 sessions the ensemble beats the base rate in log-loss (Holm p = 0.006) although its accuracy is not distinguishable from the baseline (p = 0.144), because accuracy only counts which side of 0.5 a forecast falls on.
- **Out-of-fold, only 1 session passes.** The 1-session ensemble beat the base rate (Holm p = 0.0001) but not the best single model or the simple average after correction (Holm p = 0.74 and 1.00). The 3-, 5- and 10-session ensembles did not beat the base rate out-of-fold.
- **The ensemble earns its place at 1 session.** On the test it beats the best single model and the simple average at 1 session, and the best single model at 3 sessions, after correction. At 5 and 10 sessions nothing separates the candidates.

Intervals and p-values for accuracy and coverage come from a session-block bootstrap, which resamples whole sessions so that stocks traded on the same day stay together (1,000 draws on the test, 300 out-

of-fold). The Holm correction runs over the full set of comparisons at each horizon, so a result that survives it is not the best of many looks.

Drivers

Two features that describe the whole market, the NEPSE index's 1-session return and 20-session volatility, matter most at every horizon, and fundamentals take over at 5 and 10 sessions.

Market-wide features dominate; fundamentals matter only at 5 and 10 sessions

Rise in log-loss (x 1,000) when a feature group is shuffled; HistGradientBoosting, validation folds 7 and 8

	1 session	3 sessions	5 sessions	10 sessions
NEPSE market (2)	17.8	11.5	34.1	36.4
Fundamentals (7)	3.1	4.8	20.0	31.4
Candle shape and gaps (4)	3.5	3.9	2.9	3.0
Broker concentration (13)	3.1	3.7	2.0	2.4
Returns and momentum (11)	6.9	2.4	0.9	-0.1
Liquidity and volume (9)	1.8	1.8	2.8	1.5
Volatility and range (7)	0.9	0.3	1.6	0.8
Trend and oscillators (7)	1.4	0.3	0.7	0.2

Darker means a larger loss when shuffled. Feature counts in parentheses. Development data only; the final test was not used.

grouped permutation importance · HistGradientBoosting refit on folds 7 and 8, development data only · 2 shuffles per group

These are model dependencies, not causes. Shuffling a group measures how much one HistGradientBoosting model loses without it, on the two latest validation folds, so correlated groups can share credit. Two cautions apply:

- **Market features act as a regime label.** They take the same value for every stock in a session, so a tree can use them to recognise the period. The 1-session forecast leans on them less (loss rises 0.018) than the 5- and 10-session forecasts (0.034 and 0.036).
- **Fundamentals change only quarterly.** Their weight at 10 sessions (0.031) may reflect slow differences between stocks, or the period, rather than information the market has not already priced. The ablation in `run_pipeline.sh` (`NEPSE_FEATURESET=core`) isolates this; it has not been run here.

At 1 session the signal is spread thinly: returns and momentum (0.0069), candle shape and gaps (0.0035), broker concentration (0.0031) and fundamentals (0.0031) each add a little, and the market-wide pair adds the most (0.0178). No single stock-level group dominates at any horizon.

Backtest

No long-only strategy earns a net return that differs from zero. Portfolios built on confident "up" forecasts beat random stock picks at 1, 3 and 10 sessions but lose or barely earn after costs, and the ranker's top-10 portfolio loses money at every horizon.

The backtest runs on the untouched test. At the close of a session it buys, equal-weight and up to 10 liquid names, those whose calibrated P(up) reaches the threshold. It enters at the next session's open, exits at the close of session t + h, and deploys 1/h of capital per signal session. Costs per side are 0.36% commission, 0.015% SEBON fee and 0.10% slippage, about 0.95% round trip, plus 7.5% short-term capital-gains tax on profitable trades. Signals in names that do not trade the next session are counted as unfilled (fill rate 90–97%).

Horizon (P(up) ≥ 0.60)	Trades	Mean gross per trade	Mean net per trade [95% interval]	Hit rate	Random picks, mean net	Cumulative return	Sharpe	Max drawdown
1 session	410	+0.81%	−0.20% [−0.56%, +0.20%]	41.2%	−0.65%	−27.0%	−1.31	−29.8%
3 sessions	244	+1.65%	+0.55% [−0.25%, +1.41%]	50.4%	−0.37%	+8.2%	0.72	−12.1%
5 sessions	128	+0.94%	−0.15% [−1.98%, +1.44%]	36.7%	−0.43%	−2.1%	−0.55	−5.7%
10 sessions	409	+1.64%	+0.52% [−1.01%, +2.21%]	50.4%	−0.27%	+5.2%	1.62	−3.7%

Random picks are drawn from the same sessions with the same number of names per session (300 draws). The strategy beats them at 1, 3 and 10 sessions (no random draw did as well in 300) and at 5 sessions does not (p = 0.17). Random picks lose 0.3–0.7% a trade because costs exceed the average gross return, so beating them is a low bar. Every interval for the strategy's net return includes zero.

- **Confident forecasts cannot be traded long-only.** Almost all of the confident calls behind the 70% table are "down" calls. A long-only investor can act only on "up" calls, and there are few of them: at P(up) ≥ 0.70 there were 7 signals at 1 session and none at 3, 5 or 10, so no portfolio was run at that threshold.
- **The ranker earns a gross edge that costs erase.** The top-10 portfolio makes +0.47%, +0.62%, +0.61% and +0.67% gross per trade at 1, 3, 5 and 10 sessions, and −0.53%, −0.41%, −0.43% and −0.40% net. Cumulative returns are −72.4%, −26.8%, −17.2% and −8.6%, and the net per-trade loss is significant at 1 and 3 sessions. Its quintile spread needs a short leg that NEPSE retail investors do not have.

- **Costs dominate at short horizons.** Every 1-session trade pays the full round-trip cost, which is why the 1-session portfolio, the one with the best forecasts, is the worst performer.

Experiment register and run lineage

The five native libraries that could not be installed when this study was run — XGBoost, LightGBM, CatBoost, the Transformer and the TFT — have since been run in a separate environment on a reduced panel. A file register compiled on 9 November 2026 inventories everything produced so far: 555 records, made up of 208 files inside the six execution workspaces, 33 loose files and 314 members of ZIP archives. This section records which run produced what, so that no figure is read against the wrong experiment.

Run lineage

1. **Full-panel scikit-learn baseline — this document.** `nepse_pipeline/`, `nepse_pipeline_resume/`, `nepse_pipeline.zip` and `nepse_results.zip`. Five scikit-learn classifiers and one ranker over the full 362-stock panel. Every accuracy, calibration, significance, driver and backtest figure in this document comes from this run and from no other.
2. **Early native lite work.** `nepse_native_lite_work/` holds a seven-family out-of-fold object for folds 1–8 with early ensemble and statistical outputs. Superseded by the combined run below.
3. **Native tree branch.** `nepse_native_tree_lite_run/` holds the XGBoost, LightGBM, CatBoost and logistic out-of-fold outputs, the training log, the panel, the split manifest and stage reports.
4. **Native sequence and ranker branch.** `nepse_native_seq_ranker_lite_run/` holds the Transformer, TFT-style and LambdaRank out-of-fold and final-test outputs, saved models and feature diagnostics.
5. **Combined native seven-model run.** `nepse_native_all_models_lite_run/` merges all seven native families into one set of out-of-fold predictions, with the selected ensemble, calibration, statistics, final-test scoring, saved final models, the source panel, the scripts and its own report.
6. **Standalone refits.** The CatBoost and XGBoost/LightGBM bundles preserve per-horizon refit results, logs, summary tables and training scripts from before the combined run.

The native runs are a different experiment, not an update to this one. They use a 100-stock lite panel; the results in this document use the full 362-stock panel. The model families differ and the universe differs, so rows from the two must never appear in one table without naming the run that produced each. No native figure is quoted anywhere in this document.

Execution workspaces

Workspace	Files	Stored size	Contents
`nepse_native_all_models_lite_run/`	88	158.5 MB	Latest merged native seven-model lite-panel run: out-of-fold predictions, ensemble and calibration, statistics,

Workspace	Files	Stored size	Contents
			final-test metrics, saved final models, scripts, source panel and research report.
`nepse_native_seq_ranker_lite_run/`	60	129.7 MB	Earlier Transformer / TFT-style / LambdaRank branch: sequence and ranker outputs, models, feature diagnostics, reports and scripts; also some stale process-ID marker files.
`nepse_native_tree_lite_run/`	8	67.8 MB	Earlier tree branch: XGBoost, LightGBM, CatBoost and logistic out-of-fold outputs, training log, panel, split manifest and stage reports.
`nepse_native_lite_work/`	11	130.8 MB	First combined native attempt. Its out-of-fold object already covers all seven families, but it predates the merged bundle.
`nepse_pipeline/`	21	212 KB	Source of the pipeline described in this document: ingestion, auxiliary sources, panel, audit, models, ensemble, statistics, final test, backtest export and feature importance.
`nepse_pipeline_resume/`	20	165 KB	Resumed copy of the same source tree, including the model, backtest and reporting scripts.

Packages and reports

Package or document	What it is
`nepse_native_all_models_lite_bundle.zip` (82.3 MB, 88 entries)	Complete combined native seven-model lite-panel bundle.
`nepse_native_all_models_lite_results.zip` (195 KB, 34 entries)	Compact reports, metrics and logs for the same run.
`research_report_native_all_models_lite.md` (12.8 KB)	Narrative report for the combined native run, with its own addendum file.
`nepse_native_sequence_ranker_lite_bundle.zip` (69.3 MB)	Earlier sequence and ranker bundle, with a compact results package beside it.

Package or document	What it is
`nepse_catboost_completed_bundle.zip` (24.3 KB, 17 entries)	Standalone CatBoost refit: per-horizon results, logs and the training script.
`nepse_xgb_lightgbm_run_bundle.zip` / `…_spec_refit_bundle.zip`	XGBoost and LightGBM first run and specification-aligned refit, with summary CSVs.
`nepse_results.zip` (268 KB) and `nepse_pipeline.zip` (57.5 KB)	Stage reports, figures and source for the baseline run documented here.
`NEPSE_COMPLETE_MODEL_EXECUTION_FILE_REGISTER.csv`	The register itself: 555 rows of record type, container, path, byte size and description.

What the combined native run holds

- `oof.pkl` — merged out-of-fold predictions for XGBoost, LightGBM, CatBoost, logistic regression, the Transformer, the TFT-style model and LambdaRank; `ens_oof.pkl` — ensemble and calibration outputs; `test_predictions.pkl` — final-test predictions with row and session metadata.
- `models/` — saved final models per horizon in each library's own format (`.json` for XGBoost, `.cbm` for CatBoost, `.txt` for LightGBM and the ranker, `.pt` for the Transformer and TFT, `.joblib` for logistic regression and the second-level models).
- `reports/` — stage reports, candidate metric tables for out-of-fold and final test, bootstrap and Holm results, ranker statistics and feature-group diagnostics.
- `scripts/`, `source/TRAINING_SPEC.md` and `source/nepse_model_panel_lite.csv.gz` — the code and the exact panel the run read, so the run can be repeated without rebuilding the panel.
- `splits.json`, `selection.json`, `final_test.lock`, `environment.txt`, `RUN_STATUS.json` and `CHECKSUMS.sha256` — the fold manifest, the pre-test selection and its hash, the dependency snapshot, the completion record and the integrity manifest.

Reading the register

- **Duplicates are deliberate.** The same artifact is copied into several bundles; a repeated path is a package copy, not a separate run.
- **Integrity.** The combined run's `CHECKSUMS.sha256` was verified for all 87 payload files when that bundle was built. The register covers workspaces and archives beyond that bundle, which carry no such manifest.
- **Stale markers.** The older sequence and ranker workspace still contains `.pid` files. They record process IDs from finished runs and are not restart instructions.
- **An inventory is not a result.** Listing a run does not make it complete, statistically independent of the others, or deployable; the runs share a panel and overlap heavily in their folds.

Limitations and threats to validity

The numbers in this document describe one market, one final-test window and a stand-in model family; read them as a clean baseline, not as a deployable edge.

- **Stand-in models.** CatBoost, LightGBM, XGBoost, the Transformer and the TFT never ran here, because package installs were blocked. Every result comes from scikit-learn models: HistGradientBoosting, ExtraTrees, logistic regression, an MLP, a 20-session windowed MLP and a pointwise HistGradientBoosting ranker. `FAMILY=native START=6 bash run_pipeline.sh` reruns stages 6–13 with the named libraries, and that is how the separate native lite-panel run was produced; because that run uses 100 stocks rather than 362, it does not replace the figures here.
- **Floorsheet gaps.** In 14 months (Feb 2022, Apr 2022 – Mar 2023, Mar 2024) the floorsheet holds only 20–60 stocks a session. Prices stay complete because they come from the adjusted OHLCV table, but broker and trade-size features are missing there (present for 29.2% of 2022 samples) and validation folds 3 and 4 fall almost entirely inside the gap.
- **Base rate.** Only 45.5–46.6% of labelled outcomes are up, so a model that leans toward "down" beats a coin flip without skill. Every accuracy is therefore shown against the majority-class baseline of the same window.
- **Overlapping labels.** For 3, 5 and 10 sessions, neighbouring labels share most of their return path, so the effective sample is far smaller than the row count. The 25-session embargo, the session-block bootstrap and HAC-corrected Diebold–Mariano tests account for this; a plain binomial interval would be too narrow.
- **Confidence threshold.** Thresholds of 0.55–0.70 were fixed before scoring, but four thresholds across four horizons are still sixteen looks. A 70% cut-off selects the most confident calls; it does not make the model 70% accurate overall, and its coverage is small.
- **Publication lag.** Fundamentals are used from quarter end + 45 days. That lag is an assumption, since none of the fundamentals tables records when a report was published.
- **Survivorship.** The OHLCV table lists only symbols still trading; 82 delisted ordinary shares use floorsheet prices. Companies delisted before October 2018 are absent altogether.
- **Price jumps and flat outcomes.** Residual moves above 11% (the circuit limit is 10%) were neutralised on 225 symbol-sessions, and flat outcomes (0.5–5.6% of rows by horizon) have no direction label, so they are excluded.
- **Execution.** The backtest enters at the next open with 10 bp slippage per side and no market impact. Thin names may fill worse, and the strategy is long-only because retail investors on NEPSE cannot short.

Reproducibility

The whole study reruns from the source dumps with one script, and every stage is resumable or idempotent, so an interrupted run restarts where it stopped.

```
pip install pandas numpy scipy scikit-learn joblib matplotlib lightgbm
catboost xgboost torch
NEPSE_UPLOADS=/path/to/zips_and_sql NEPSE_AUX=/path/to/fundamental_sql
NEPSE_WORK=./work \
  FAMILY=native bash run_pipeline.sh        # FAMILY=sklearn reproduces the
```

```
numbers in this document
START=6 FAMILY=native bash run_pipeline.sh  # resume at the models;
START=<stage> works for any stage
```

The cleaned panel, splits and audit (stages 1–5) are independent of the model family, so a native run can copy them across and begin at stage 6. `NEPSE_FEATURESET=core` drops the seven fundamentals for the ablation; run it in its own work folder.

Results here were produced with Python 3.13.18, pandas 3.0.5, numpy 2.5.3, scikit-learn 1.9.1, scipy 1.18.1, matplotlib 3.11.2 and joblib 1.6.0 on 2 CPUs. Random seeds are fixed in `common.py`, and the final-test selection is hashed before the test is scored.

Source file	SHA-256
`floorsheet_data_part1.zip`	`2f0d17037a60c3a032ce7b572dbc6874dba23f2a4974c2ae9cacf37f42011595`
`floorsheet_data_part2.zip`	`f1dc39c48ca63786b75ee6fedc594590315a5e0ffc1ea2ef54b1bec060039365`
`floorsheet_data_part3.zip`	`25f403e6889a11c6c19371a8fbf0d7a427cfab0ee38296b2edb130d7794583fc`
`ohlcv_data.sql`	`2de9ccd63a6703d7a041bf063027c78498f7a642f21688b516d4097c77b3fdc5`
`nepse_prediction_featurestore.sql`	`3c86d6177c5d80c1aa70660143abbbe9fdb5c9a6852df1bbf3a58a4ce4627e6c`
`fundamental_data.sql`	`babd2dd157d735cfc6da4cd38ef9f0ae2e4292a034d06af9af8a04ea10fabf5e`

Every run is inventoried in the 9 November 2026 execution register (`NEPSE_COMPLETE_MODEL_EXECUTION_FILE_REGISTER.csv`, 555 rows), which names each workspace, bundle and archive member, its size and its role. Each bundle carries its own `environment.txt` and, for the combined native run, a verified `CHECKSUMS.sha256`.

Each stage writes a JSON report to `$NEPSE_WORK/reports/`, and stage 13 assembles them into `research_report.md` together with the figures and the model manifest.